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Smoke and Ash Effects on Seedling Emergence from Germinable Soil Seed Bank in Fescue Prairie[☆]



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ABSTRACT

Seedling recruitment plays a crucial role in recovering plant communities after disturbance. As a natural disturbance, fire can mediate species composition in fescue prairie. However, little was known about the effects of important fire cues on seedling recruitment in fescue prairie. Soil seed bank samples were taken from the top 5 cm of the soil profile and separated into litter, 0- to 1-cm, and 1- to 5-cm layers. Seedlings emerging from soil seed banks incubated in the greenhouse were examined after applying smoke, ash, and smoke plus ash in 2013 and 2014, to assess their effects on the density, richness, and composition of seedlings emerging from the soil seed bank in fescue prairie. Smoke plus ash significantly increased the number of *Artemisia ludoviciana* Nutt. seedlings emerging from 0- to 1-cm soil layer and *Conyza canadensis* (L.) Cronquist seedlings emerging from the litter layer ($P < 0.05$), while ash significantly increased the number of *Artemisia frigida* Willd. seedlings emerging from 0- to 1-cm soil layer ($P < 0.05$). Densities of total seedlings emerging from the 0- to 1-cm soil layer were increased by smoke plus ash in 2013 and by ash in 2014 ($P < 0.05$). Smoke plus ash and ash alone had more prominent effects on seedling density and richness of native forbs. Species composition was altered by ash in the 0- to 1-cm, 0- to 5-cm, and all layers combined in 2013 ($P < 0.05$). Direct fire cues appear to stimulate recruitment of some species, especially native forbs, contributing to potential changes in species composition of fescue prairie.

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Introduction

Fescue prairie is one of the most threatened ecosystems in Canada, located in central and southwestern Alberta and west central Saskatchewan (Moss and Campbell 1947; Coupland and Brayshaw 1953). Due to urbanization and cultivation, 95% and 85% of fescue prairie in Saskatchewan and Alberta has been severely degraded, respectively. In addition, fire suppression leaves fescue prairie susceptible to invasion by shrubs (Anderson and Bailey 1980). Reintroducing fire is critical in maintaining diversity in remnant fescue prairie (Romo 2003), but effects of fire on plant community composition in fescue prairie vary with burning season (Bailey and Anderson 1978; Redmann et al. 1993) and fire frequency (Anderson and Bailey 1980). Species composition shifted in favor of perennial forbs for at least 3 years after burning in a fescue prairie in Alberta, Canada (Bailey and Anderson 1978). High germination and seedling establishment occurred immediately after burning (Keeley et al. 2012).

Soil seed bank is regarded as a biodiversity reservoir in restoring degraded or invaded ecosystems (Thompson and Grime 1979). Romo and Gross (2011) concluded that species composition of

seedlings that emerged from the soil seed bank in fescue prairie was affected by burning history and season of burning. Although seedling recruitment from the soil seed bank is fundamental to predicting plant species composition in response to disturbances, only limited attention has been given to the effects of burning on seedling emergence from the soil seed bank in fescue prairie.

Burning alters soil nutrients, light availability, temperature fluctuations on the soil surface, and competitiveness among species (Keeley et al. 1985; Auld and Bradstock 1996). Altered environmental conditions account partly for the surge in regeneration from soil or canopy-stored seed bank after burning. In addition, solitary or combinations of direct fire cues, such as heat shock (Keeley et al. 1985), plant-derived smoke (Roche et al. 1997; Abella 2009), and ash (Gonzalez-Rabanal and Casal 1995; Izhaki et al. 2000) have been reported to exert promotive, inhibitive, or neutral effects on seedling emergence from soil seed banks. Smoke can break seed dormancy and stimulate seed germination (De Lange and Boucher 1990; Keeley and Fotheringham 1997) and has been shown to increase seedling densities and richness of seedlings emerging from the soil seed bank in mesic grasslands in South Africa (Ghebrehiwot et al. 2012). With increasing soil depth, the extracted content of 3-methyl-2H-furo[2,3-c]pyran-2-one (KAR₁), the major active compound in smoke, decreased (Ghebrehiwot et al. 2011). Smoke also increases germination of species native to semiarid grasslands in North America (Schwilk and Zavala 2012). Inhibiting effects of ash on seed germination and seedling establishment have been reported in

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herbaceous (Sweeney 1956), shrub (Gonzalez-Rabanal and Casal 1995), and tree species (Izhaki et al. 2000). However, germination of four montane species in Mexico was positively affected by ash (Zuloaga-Aguilar et al. 2011). The combination of fire cues, specifically heat and smoke, can influence germination independently or synergistically. However, the interactive effect of smoke and ash on seedling emergence from the soil seed bank remains relatively unknown.

Despite the importance of the seed bank in vegetation dynamics, roles of different fire cues in general, and smoke and ash specifically, on seedling emergence from the soil seed bank in fescue prairie are poorly understood. The objective of this study was to determine the effects of smoke, ash, and their combination on the density and composition of seedlings emerged from the soil seed bank in fescue prairie. The following hypotheses were tested: 1) smoke, ash, and smoke plus ash alter species composition and promote seedling emergence from the soil seed bank in fescue prairie; 2) seedlings emerging from upper soil layers of the soil seed bank respond stronger to smoke, ash, and smoke plus ash; and 3) smoke, ash, and smoke plus ash favor seedling density and species richness of forbs and native species emerging from the soil seed bank.

Material and Methods

Study Site

Field work was conducted at Kernen Prairie, an approximately 130-ha parcel near Saskatoon, SK, Canada (52°10'N, 106°33'W, elevation 510 m) in 2013 and 2014. The long-term annual temperature averages 2.2°C (Environment Canada 2013), while the long-term average temperatures in May, June, July, and August (growing season) are 11.5°C, 16.0°C, 18.2°C, and 17.3°C, respectively (Environment Canada 2013). The long-term annual precipitation averages 350 mm, with more than half of it occurring from May to August (Environment Canada 2013). In 2013, temperatures averaged 13.0°C in May, 15.5°C in June, 17.4°C in July, and 18.9°C in August. Total precipitation was 15.2 mm in May, 115.9 mm in June, 35.2 mm in July, and 14.7 mm in August (Environment Canada 2013). In 2014, temperatures averaged 10.1°C in May, 14.1°C in June, 18.3°C in July, and 17.9°C in August. Total precipitation was 61.1 mm in May, 94.8 mm in June, 44.5 mm in July, and 18.5 mm in August (Environment Canada 2014). Kernen Prairie is dominated by *Festuca hallii* Vasey and other C3 plants (Coupland and Brayshaw 1953). Soils in Kernen Prairie are Bradwell and Sutherland Orthic Dark Brown Chernozems (Acton and Ellis 1978). The growing season has approximately 110 frost-free days.

Experimental Design

Fifty 3 × 3 m experimental plots were established on 17 April 2012 for this study. Twenty-five plots were used for study in 2013, and another 25 plots were used for study in 2014. Four subplots of the same size (25 × 25 cm) were randomly selected within each plot. Twenty-five 25-cm² soil core samples were collected from the top 5 cm of the soil profile in each subplot using iron cylinders with diameter of 5.6 cm and were subjected to one of four treatments: control, smoke, ash, and smoke plus ash. Soil samples were collected on 4 May 2013 and 10 May 2014. Except for the control, soil cores were collected after clipping the phytomass to the ground in each subplot. Clipped phytomass was collected and dried in an 80°C oven for 2 days to determine dry weight. Dry weight averaged at 87 ± 4.9, 94 ± 7.2, and 99 ± 8.0 g for smoke, ash, and smoke plus ash treatments in 2013, respectively. It averaged at 74 ± 4.5, 73 ± 5.9, and 75 ± 6.7 g for smoke, ash, and smoke plus ash treatments in 2014, respectively. Phytomass from each subplot was then combusted to generate smoke solution or ash. The apparatus to produce smoke solutions and ash was constructed from a wooden board fixed with an electric ring heater. A sufficient gap was left under the board for the power cord of the ring heater to exit. A metal, 75-L

garbage can with two holes drilled on either side was inverted over the heater. Pressure gauges connecting an air-supplying hose on one side and a silicon tube on the other side were affixed to the two holes. A pot for holding plant material and two weights to hold the garbage can in place completed the apparatus. Each smoke, ash, or smoke plus ash sample was produced by smouldering phytomass that was put into a container and placed on the electric ring heater. The garbage can was inverted over the wooden board enclosing the ring heater and the metal container containing plant materials. The weights were placed on top of the garbage can to eliminate smoke leakage. Air was forced into the combustion chamber at a pressure of 70 – 100 kPa. Smoke produced from the samples was continuously passed through the silicon tube and bubbled into 500 mL of distilled water in a water bottle to make smoke solutions. Ash was left in the container after combustion. In total, 25 smoke solution samples, 25 ash samples, and 25 smoke plus ash samples were produced within each year in this way.

Each soil core sample collected from the top 5 cm of the soil profile was divided into the litter layer on the soil surface, 0–1 cm, and 1–5 cm depths. Common layers for the 25 soil cores from each subplot were combined, mixed, and spread onto plastic trays (52 × 26 × 7 cm) to < 1 cm in depth. Root fragments, rhizomes, and plant materials were removed. The litter layer was spread on a 1-cm layer of sterile sand placed at the bottom of plastic trays measuring 26 × 26 × 6 cm. Trays containing soil cores from the same plot were randomly placed as blocks in a greenhouse. The air temperature during the growing season in the greenhouse averaged 27 ± 3°C during the day and 21 ± 4°C at night. Natural light was supplemented with two 400-W, high-pressure sodium lights (18-h photoperiod, average of 600 μmol · m⁻² · s⁻¹) for each bench. Each of the 25 blocks had four treatments including soil cores without any treatment, soil cores treated with smoke, soil cores treated with ash, and soil cores treated with smoke plus ash.

The amount of smoke, ash, or smoke plus ash applied to the litter layer and two soil layers was based on the volumetric ratio among these three layers. More specifically, we calculated the volumes of 0- to 1-cm and 1- to 5-cm soil layers of 25 soil cores collected from same subplot, which was 625 cm³ (25 × 25 cm² × 1 cm) and 2 500 cm³ (25 × 25 cm² × 4 cm), respectively, in both years. We used volumetric cylinder to measure the litter volume of 25 soil cores collected from the same subplot. In total, we had 100 (25 × 4) subplots in each year. The litter volume within each subplot averaged at 220 and 270 mL in 2013 and 2014, respectively. Hence, the ratios for applying smoke, ash, or smoke plus ash obtained from the phytomass collected from a specific subplot to the litter, 0- to 1-cm, and 1- to 5-cm soil layers collected from the corresponding subplot were 220: 625: 2 500 and 270: 625: 2 500 in 2013 and 2014, respectively. Water was applied immediately after applying different treatments. All trays were watered manually to field capacity every day. Extra water was drained through small holes at the bottom of each tray. Seedling emergence was recorded weekly for 14 weeks, and new seedlings were transplanted in pots until they were identified. In total, data from five layers (litter layer, 0- to 1-cm depth, 1- to 5-cm depth, 0- to 5-cm depth) and all layers combined (litter layer plus 0- to 5-cm depth soil layer) were analyzed.

Data Analysis

Analysis was conducted on individual species' seedling densities, total seedling densities, species richness, seedling densities and species richness of functional groups of forbs, graminoids, native, and non-native plants, species diversity (Diversity indices), and seedling emergence rate from the soil seed bank in the greenhouse. Data were square root, log, or log (x + 1) transformed and subjected to analysis of variance (ANOVA) in a split plot in a complete randomized design with 25 replicates. Transformations were applied to meet normality assumptions. Twenty-five plots within each year were the whole plot experimental unit, with year as the whole-plot factor, while each plot within each year was the subplot experimental unit, with different treatments

(smoke, ash, and smoke plus ash) as subplot factors. When the interaction of year $\times$ treatment was significant ($P \leq 0.05$), data were analyzed within years. Significantly different treatment means were separated using a Tukey test at $P \leq 0.05$. The mixed model procedure in SAS version 9.3 was used for data analyses (SAS Institute Inc., Cary, NC), with main effects and possible interactions of year and treatments as fixed factors. Block (the whole-plot unit) was factored into the model as a random effect.

Differences in species composition among treatments were tested using a PerMANOVA test, a permutation-based multivariate analysis of variance in PC-ORD version 6.0 (McCune and Grace 2002). PerMANOVA controls for variation contributed by a blocking factor, and data were tested using 999 random permutations. Densities of seedlings emerging for individual species in each plot were relativized by dividing densities by the total seedlings emerging in each plot. Sorensen distances were used to express similarity of species composition among treatments (McCune and Grace 2002). Blocked indicator species analysis was used to determine the frequency and abundance of species emerging from the soil seed bank for specific treatments (Dufrene and Legendre, 1997). Seedling densities emerging for individual species in each plot were relativized by being dividing by the total number of seedlings emerging from each corresponding block (McCune and Grace 2002). A Monte Carlo simulation of 10 000 runs was used to evaluate the statistical significance of indicator values (McCune and Grace 2002).

Results

Seedling Density

A total of 21 434 and 32 895 seedlings were identified from the soil seed bank samples in a 6.25 m² sampling area in 2013 and 2014,

respectively. Across all treatments and control, seedling numbers of native species were 1.5-fold and 1.7-fold greater than those of non-native species in 2013 and 2014, respectively. Forb seedling numbers were 1.3-fold greater than those of graminoids in both years. The percentage of unidentified seedlings was $< 1\%$. *Androsace septentrionalis* L. was the most abundant forb emerging (73% and 69% of total forbs in 2013 and 2014, respectively) while *Poa pratensis* L. was the most abundant graminoid emerging (84% and 81% of total graminoids in 2013 and 2014, respectively).

In the control plots, 81% of *Artemisia frigida* Willd. seedlings and 75% of *Artemisia ludoviciana* Nutt. seedlings from the 0- to 5-cm soil layer emerged from the 0- to 1-cm soil layer in both years. The ash treatment increased densities of *A. frigida* emerging from the 0- to 1-cm soil layer by 110% compared with the control (Fig. 1). Densities of *A. ludoviciana* emerging from the 0- to 1-cm soil layer were significantly increased by smoke plus ash by 253% compared with the control. Densities of *Conyza canadensis* (L.) Cronquist emerging from the litter layer and all layers combined were 2.6-fold and 1.7-fold greater in smoke plus ash than those in the control, respectively.

Total densities of seedlings emerging from the litter layer were increased by ash in both years (Fig. 2). Total densities of seedlings emerging from the 0- to 1-cm soil layer and all layers combined were affected by the interaction of treatment and year and were significantly increased by smoke plus ash in 2013 and by ash in 2014 as compared with the control. Smoke reduced densities of total seedlings emerging from the 0- to 1-cm soil layer in 2013.

Densities of seedlings from the various functional groups emerging from different soil layers in the soil seed bank responded differently to smoke, ash, and smoke plus ash (Fig. 3). In the control, 60% of native species or forbs emerging from the soil seed bank emerged from the

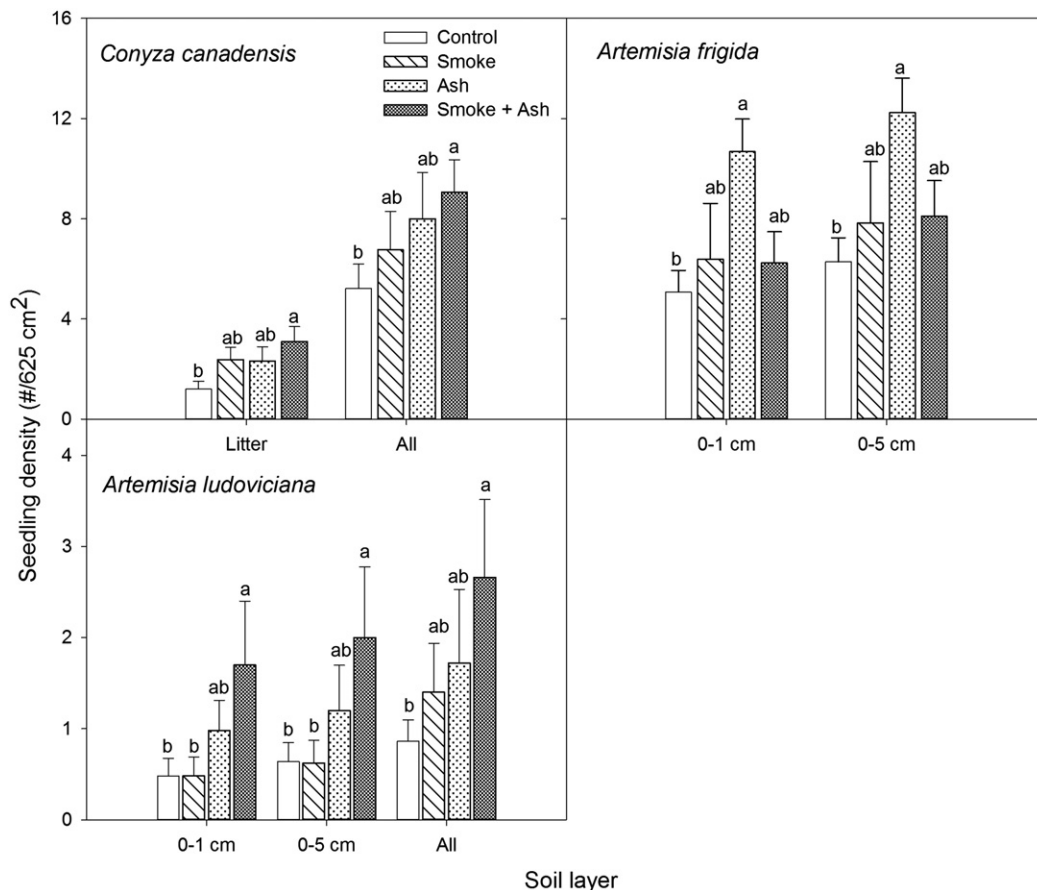


Figure 1. Densities of species in which total seedling densities emerging from the soil seed bank were significantly affected by smoke, ash, or smoke plus ash in 2013 and 2014 combined. Values represent means $\pm$ SE of 5 replicates. Means with different letters within soil layers were significantly different ($P \leq 0.05$).

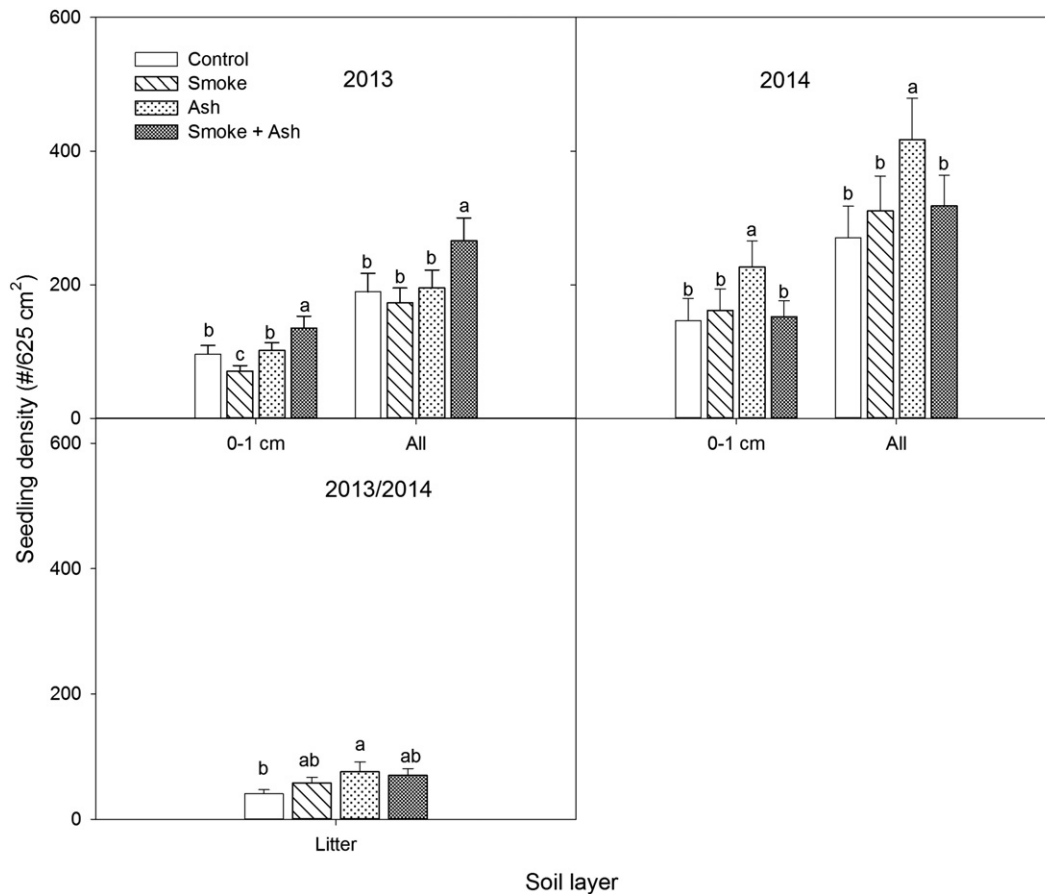


Figure 2. Total seedling densities emerging from the soil seed bank as affected by smoke, ash, and smoke plus ash in 2013 and 2014. Values represent means $\pm$ SE. Means with different letters within soil layers were significantly different ($P \leq 0.05$).

0- to 1-cm soil layer in both years. Densities of native species in the 0- to 1-, 0- to 5-cm, and all layers combined were affected by the interaction of treatment and year. Smoke plus ash increased densities of native species emerging from the 0- to 5-cm soil layer in 2013. In 2014, densities of native species responded positively to ash in the 0- to 1-cm soil layer, with a 67% increase in their abundance. Ash increased densities of forbs in the 0- to 1-cm and all layers combined by 44% and 34%, respectively, as compared with the control. Smoke, ash, or smoke plus ash did not significantly affect densities of non-native species and graminoid seedlings emerging from the soil seed bank.

Species Composition

Smoke significantly altered species composition as compared with the control in the litter layer in 2013 (Table 1). Species composition in the ash treatment was significantly different from that in the control in the 0- to 1-cm and 0- to 5-cm soil layers in 2013. As compared with the control, none of the three treatments significantly altered species composition in any soil layers in 2014.

Species richness of total seedlings emerging from the soil seed bank responded consistently to treatment effects in both years (Fig. 4A). Species richness of total seedlings emerging from all soil layers in the soil seed bank was significantly increased by smoke plus ash as compared with the control. Except for the 1- to 5-cm and 0- to 5-cm soil layers, species richness of total seedlings from all soil layers in the soil seed bank was significantly increased by ash as compared with the control. Smoke increased species richness of total seedlings emerging from the litter layer.

Effects of smoke, ash, and smoke plus ash on species richness of native species varied among different soil layers (Fig. 4B). Species richness

of native species was enhanced by smoke plus ash in all soil layers as compared with the control. Ash increased species richness of native species emerging from the litter and 0- to 1-cm soil layer. Species richness of native species in the litter layer was increased by smoke.

Species richness of forbs and graminoids responded differently to smoke, ash, and smoke plus ash in different soil layers, with forbs being more positively affected (Fig. 4C). Species richness of forbs was significantly increased in all soil layers by smoke plus ash. Ash increased species richness of forbs emerging from the litter and 0- to 1-cm soil layers. Species richness of graminoids emerging from all soil layers did not vary significantly among treatments. Smoke, ash, and smoke plus ash did not affect species diversity and evenness of seedlings emerging from different soil layers in the soil seed bank (data not shown).

Rate of Seedling Emergence

The rate of seedling emergence of total seedlings and functional groups emerging from different soil layers in the soil seed bank only responded to smoke, ash, or smoke plus ash in 2013 (Fig. 5). Total seedlings emerged faster in ash and smoke plus ash in the litter and all layers combined. Total seedlings emerged faster in the litter layer but more slowly in the 0- to 1-cm and 0- to 5-cm soil layers in smoke as compared with the control. Seedlings of forbs and native species emerged faster in smoke plus ash in the litter, 0- to 5-cm, and all layers combined. Smoke plus ash increased the rate of emergence of forbs and native species in the litter layer. Seedlings of forbs in the 0- to 1-cm and 0- to 5-cm soil layers and seedlings of native species in the 0- to 1-cm soil layer emerged more slowly in smoke. Seedlings of graminoids emerged faster in smoke, ash, and smoke plus ash in the litter layer. Seedlings of graminoids emerged more slowly in smoke in the 0- to 1-cm soil layer

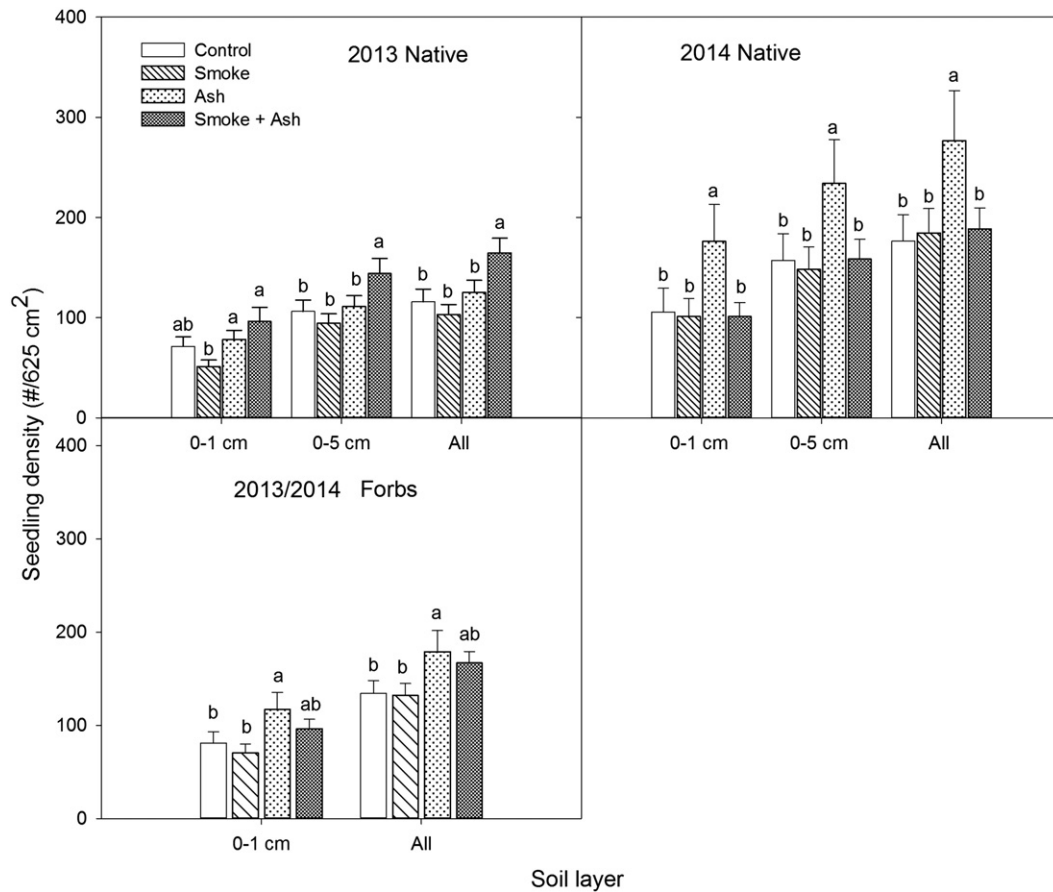


Figure 3. Densities of native species and forb seedlings emerging from different soil layers in the soil seed bank as affected by smoke, ash, and smoke plus ash in 2013 and 2014. Bars represent means $\pm$ SE of 25 replicates. Means with different letters for native species or forbs within soil layers were significantly different ($P \leq 0.05$).

but faster in smoke plus ash in all layers combined. Seedlings of non-native species emerged faster in ash and smoke plus ash in the litter layer but more slowly in smoke in the 0- to 1-cm soil layer.

Discussion

Ash and smoke plus ash had positive effects on the density, richness, and emergence rate of seedlings from the soil seed bank. Contrary to our hypothesis, smoke solutions had few positive effects on density, richness, and emergence rate of seedlings, which may be due to the relative high smoke concentration applied. Consistent with our hypothesis, density, richness, and emergence rate of functional groups responded differently to treatments, with forbs and native species being affected more profoundly.

Table 1

Pairwise comparisons from multivariate permutational analysis (PERMANOVA) of differences in species composition among the control, smoke, ash, and smoke plus ash treatments in different soil layers in soil seed bank in 2013 and 2014

Comparison	Soil layer	2013		2014	
		t value	P value	t value	P value
Control vs. smoke	Litter	1.57	0.01	0.75	0.84
Control vs. ash	0-1 cm	1.63	0.04	0.81	0.64
Smoke vs. ash	0-1 cm	1.63	0.05	1.20	0.28
Control vs. ash	0-5 cm	1.57	0.04	0.84	0.57
Smoke vs. ash	0-5 cm	1.65	0.04	0.91	0.40
Smoke vs. ash	All layer	1.63	0.04	1.00	0.43

Effects of Smoke and Ash on Seedling Emergence from Soil Seed Bank

The number of seedlings emerging from the soil seed bank varied between 2013 and 2014. Densities of seedlings emerging from the soil seed bank are subjected to variation in seed production, seed viability, and seed dormancy status in each year (Baskin and Baskin 1998). In addition, soil cores were collected between the times of seed dispersal and germination in this study, which included both transient and persistent seed banks (Baskin and Baskin 1998). Seedlings emerging from transient soil seed banks are more subjected to temporal variability (Thompson and Grime 1979).

The mechanism by which smoke promotes seed germination has not been determined, likely because it varies among species (Keeley and Fotheringham 1997). One possible mechanism of smoke in breaking seed dormancy and promoting seed germination is by changing gibberellic acid (GA) and abscisic acid (ABA) metabolism (Gardner et al. 2001; Nelson et al. 2009). Contrary to the well-reported promotional effects of smoke on seedling emergence from soil seed banks in various ecosystems (Keeley and Fotheringham 1997; Ghebrehewot et al. 2012), smoke had a neutral or negative effects on densities and emergence rates of total seedlings and functional groups emerging from the soil seed bank in our study. The highly concentrated smoke solutions applied in our study are likely to be the reason for the observed responses. Diluted smoke solutions made from *F. hallii* have been shown to stimulate germination of native species in fescue prairie, including *A. frigida*, *A. ludoviciana*, and *C. canadensis* (unpublished data). High concentrations of aqueous smoke solutions can also reduce germination due to the presence of inhibiting or toxic compounds within concentrated smoke solutions (Dixon et al. 1995). Diluted smoke solutions have been used in most studies in which seedling emergence from the soil

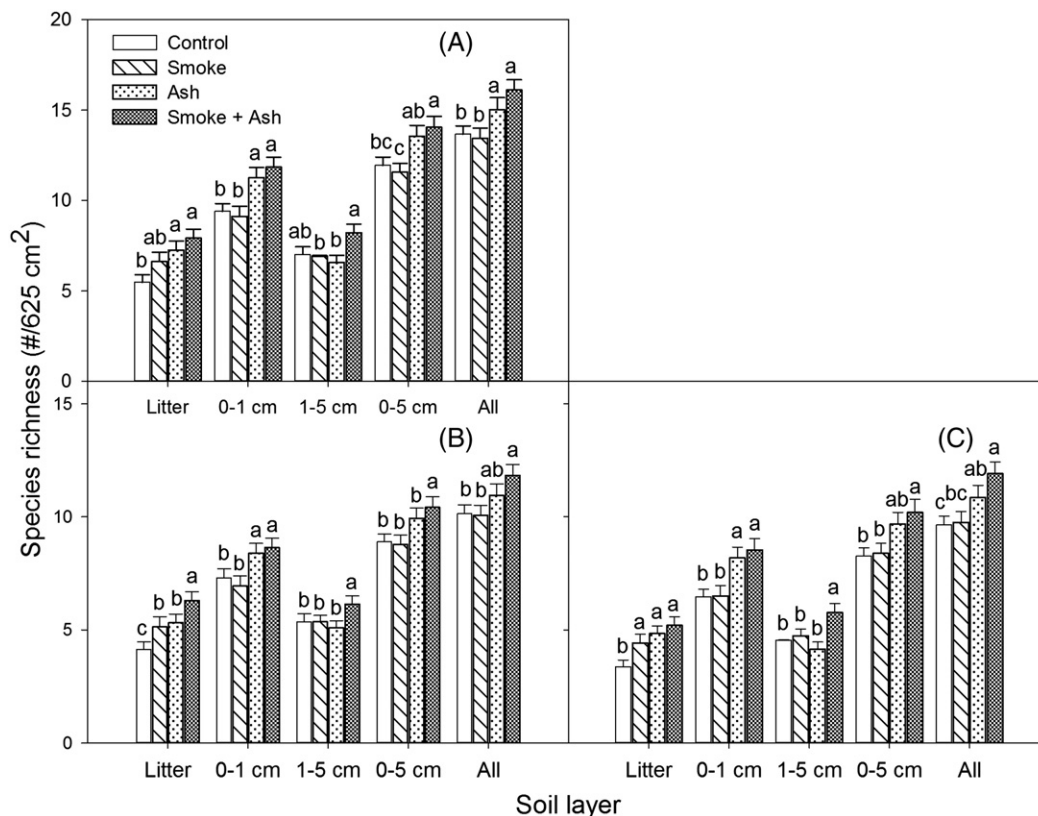


Figure 4. Species richness for total seedlings (A), native species (B), and forbs (C) emerging from different soil layers in the soil seed bank as affected by smoke, ash, and smoke plus ash in 2013 and 2014 combined. Bars represent means $\pm$ SE of 25 replicates. Means with different letters within soil layers were significantly different at $P \leq 0.05$.

seed bank responded positively to smoke (Abella 2009; Ghebrehwot et al. 2012). In addition, different methods used to apply smoke solutions may contribute variation in responses. For example, Ghebrehwot et al. (2012) applied diluted smoke solutions to the soil seed bank weekly during the whole experimental period. Enright et al. (1997) used aqueous smoke solutions to water soil samples every 2 days for the first 4 weeks. In our study, to mimic field conditions, smoke solutions were applied to the soil seed bank only once, and then tap water was used for watering to mimic precipitation after burning. It is worth noting that the amount of smoke being trapped using the combustion chamber was much greater than that encountered in the field during burning.

Ash had positive effects on densities of *A. frigida* seedlings emerging from the soil seed bank. Izhaki et al. (2000) attributed the effects of ash on germination and seedling establishment to blocked light and altered soil chemistry. In our study, approximately 27 and 21 g of ash were divided and applied to the litter, 0- to 1-cm, and 1- to 5-cm soil layer on the basis of volumetric ratio among these three layers in each replicate for ash treatment in 2013 and 2014, respectively. The amount of ash we applied barely covered the surface of trays used in our tests, thus avoiding its light blocking effect. The high pH of ash, together with soil and species specific germination characteristics, can also affect seed germination (Henig-Sever et al. 1996) and seedling establishment (Ne'eman and Izhaki, 1998). For example, it was reported that *Pinus halepensis* Mill., *Cistus salviifolius* L. and *Cistus creticus* (L.) Heywood, the three dominant plant species in native forests in Israel, were all sensitive to high pH (Henig-Sever et al., 1996; Izhaki et al. 2000). Ash decreased seedling emergence of these three species because it increased soil pH from 7.0 to 8.5 (Henig-Sever et al., 1996). In our study, soils are Bradwell and Sutherland Orthic Dark Brown Chernozems with a pH of 5.4 (Curtin and Ukrainetz 1997). Alternatively, the high pH of the ash can neutralize acidic soil, creating more suitable germination conditions for certain species. The optimum pH for germination of *A. frigida* is between 5.8 and 7.0 (Wilson 1982). Unfortunately, we did not measure

pH because of the potential impact of destructive soil sampling on seedling emergence.

Smoke plus ash but not smoke nor ash alone increased seedling densities of *A. ludoviciana* and *C. canadensis* from the soil seed bank. This suggests that combinations of fire cues can positively or negatively affect seed germination compared with a single cue (Keeley and Fotheringham 2000). Figueroa and Cavieres (2012) reported that the germination of *Aphanes arvensis* L. did not respond to heat or smoke but was significantly reduced by smoke plus heat. Similarly, germination of three common shrub species in southeastern Australia was only increased by the combination of added smoke and heat (Thomas et al. 2007). Germination of *A. ludoviciana* was favored by neutral to alkaline conditions and can be stimulated by highly concentrated smoke solutions (Lara Vanessa et al. 2014), which favors the notion that smoke and ash work synergistically on *A. ludoviciana*. *C. canadensis* germinates better in a neutral-to-alkaline soil than acidic soils (Nandula et al. 2006). Although the possible interactive effect of smoke and ash could be attributed to variation in the stimulating effects of smoke on germination in different pH conditions (Keeley and Fotheringham 1998), the underlying mechanisms are still unclear and further study is necessary. The positive responses of densities of *A. frigida*, *A. ludoviciana*, and *C. canadensis* seedlings from the soil seed bank to ash and smoke plus ash indicate that direct fire cues encourage regeneration from seeds for these three species. This partly accounts for the increased abundance and canopy cover of these three species in fescue prairie and tallgrass prairie after burning (Coupland and Brayshaw 1953; Bailey and Anderson 1978; Anderson and Bailey 1980; Collins 1987).

Responses of Functional Groups to Smoke and Ash

Seedling densities for different functional groups responded differently to smoke, ash, and smoke plus ash in our 2-yr study. Smoke plus ash and ash increased seedling densities of native species from the soil

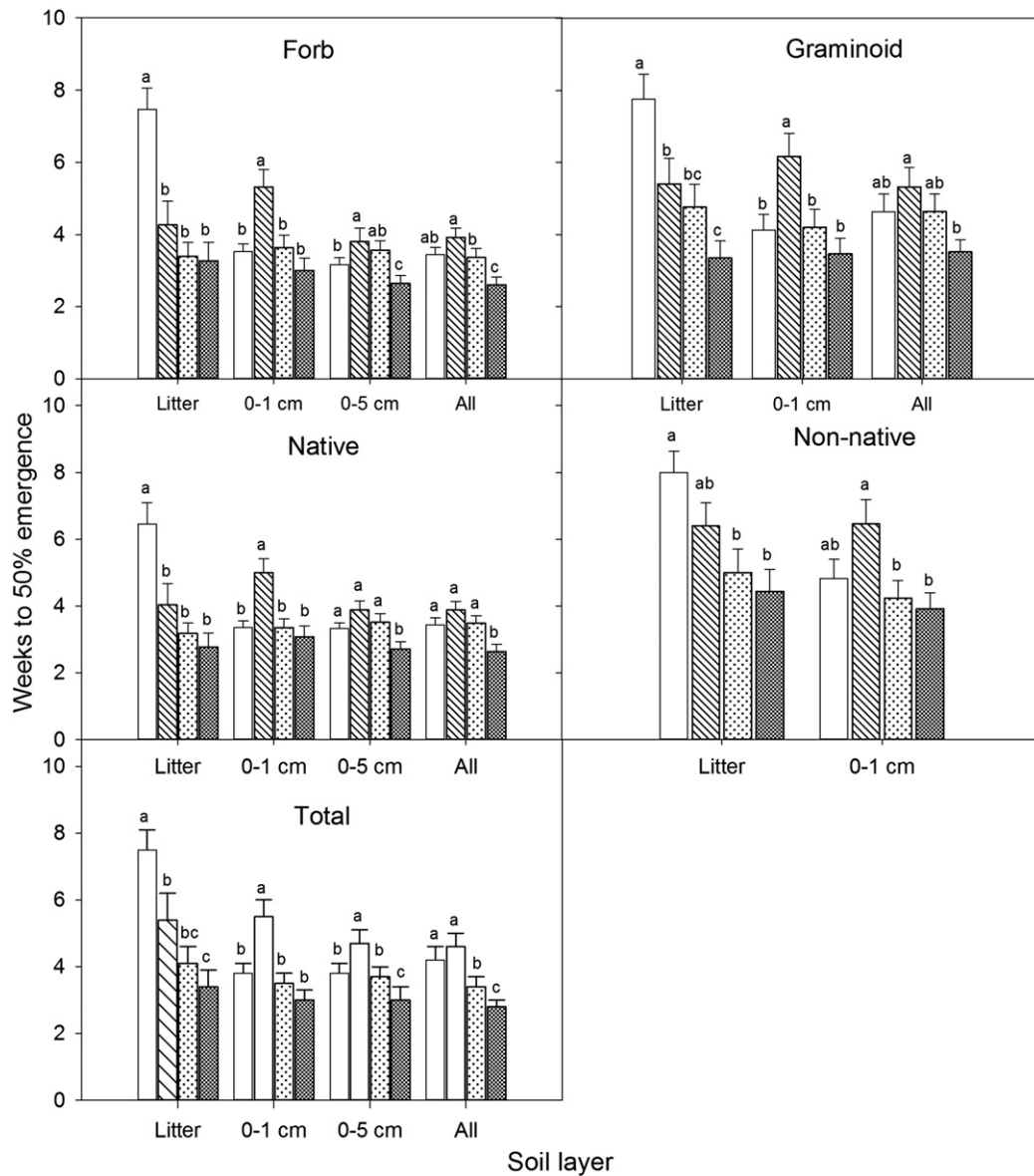


Figure 5. Weeks to 50% emergence of functional groups and total seedlings emerging from different soil layers in the soil seed bank as affected by smoke, ash, or smoke plus ash in 2013. Bars represent means $\pm$ SE of 25 replicates. Means with different letters within soil layers were significantly different ($P \leq 0.05$).

seed bank in 2013 and 2014, respectively. Increased seedling densities of native species were due to the positive responses of seedling densities of native forbs to treatments. Native and non-native graminoids emerging from the soil seed bank showed no response to treatments in both years, agreeing with the observation that seedling emergence of Poaceae was insensitive to ash (Gonzalez-Rabanal and Casal 1995). Nevertheless, ash increased forb seedling densities in the soil seed bank in both years and seedling densities of native and non-native forbs were positively affected. Species composition shifted in favor of perennial forbs, regardless of the time of burning of fescue prairie (Bailey and Anderson 1978). The positive responses of native and non-native forb seedling densities to ash indicate that elevated soil pH after burning is perhaps one of the reasons for the shift in species composition in favor of forbs in fescue prairie.

Species richness of different functional groups responded consistently to different treatments in both years. Increased number of native forb species after smoke, ash, and smoke plus ash accounted for the increased richness of native species from different soil layers in the soil seed bank. Smoke plus ash was the most effective treatment in increasing species richness for functional groups as compared with the control

in both years. Native forbs, including *Astragalus flexuosus* Douglas ex G. Don, *Draba nemorosa* L., *Stellaria longipes* Goldie, and non-native forbs including *Chenopodium album* L., *Gnaphalium uliginosum* L., and *Axyris amaranthoides* L. emerged from the soil seed bank after applying smoke plus ash but not in the control in the 2-yr study. Interestingly, all three non-native forbs are annuals, indicating fire-related cues may favor seedling emergence of species with short life cycles for non-native forbs in fescue prairie. Forbs, but not graminoids, were positively affected by smoke, ash, or smoke plus ash, further supporting the notion that direct fire-related cues might be partly attributed to increased forb abundance after burning.

Emergence rate of functional groups and total emerged seedlings responded to the treatments in 2013 but not 2014. Different treatment responses between years may be due to the fact that the amount of fuel used making smoke, ash, and smoke plus ash treatments differed between years. In 2013, emergence rates of forbs, graminoids, native species, and non-native species emerging from the soil seed bank were all affected by treatments. Seedling number of *P. pratensis*, a non-native graminoid, accounted for 84% and 94% of the total seedlings among graminoids and non-native species, respectively. Thus, observed

responses of emergence rate of graminoids and non-native species to treatments are mainly due to the presence of *P. pratensis*. Observed responses of the emergence rate of forbs and native species to treatments were most likely due to the presence of native forbs, but not to non-native forbs. Smoke plus ash increased the seedling emergence rate for native forbs emerging from the soil seed bank, further demonstrating the importance of direct fire cues in shaping species composition in favor of forbs after burning, as early recruits were being favored in a low-competition habitat.

Seedlings Emerging from Different Soil Layers Responded Differently to Smoke and Ash

Seedling densities from the litter and 0- to 1-cm soil layers responded more to the independent and interactive effect of smoke and ash, as compared with those from the 1- to 5-cm soil layer. This may be due to the uneven distribution of seeds at different soil profile depths with more viable seeds occupying the top soil profile (Izhaki et al. 2000; Ghebrehewot et al. 2012). For example, most seedlings emerged from the litter and 0- to 1-cm soil layer. Only 11% and 18% of total seedlings of *A. frigida* and *A. ludoviciana* emerged from the 1- to 5-cm soil layer in 2013 and 2014, respectively, in the control. Lack of response of seedlings of *A. frigida* and *A. ludoviciana* emerging from the 1- to 5-cm soil layer in treatments may be due to their low seedling counts, which may be under the detectable threshold. This notion is supported by Bargmann et al. (2014), who claimed that low seedling counts may account for the lack of response of herbs to smoke and heat treatments in Norwegian heathlands.

Total seedling densities in the control treatment were less in the 1- to 5-cm layer than in the 0- to 1-cm layer but three times greater than those in the litter layer in both years, indicating that total numbers of seedlings in the 1- to 5-cm soil layer was enough to detect treatment effects. Seed dormancy status of buried seeds varies among soil depths (Fukumi and Nakata 2008). Species have their own optimal depth for seedling emergence and survival. Below this optimum depth, secondary dormancy can be induced (Banting 1966). However, in our study, different dormancy status for seeds buried in different soil layers was not tested. Seed dormancy conditions would need further study to determine whether and how it can be used to explain the lack of treatment response in total seedling emergence from a 1- to 5-cm soil layer.

It should be noted that in our study, smoke solutions and ash were applied to different soil layers on the basis of the volumetric ratio among them. However, in reality, smoke and ash distribute unevenly among soil profiles, with a majority accumulating in top soil profiles. This may lead to a pattern of decreasing effectiveness of treatments on seedling emergence from a soil seed bank with increasing soil depth.

Management Implications

Ash and smoke plus ash can be used to enhance seedling emergence for rangeland restoration as evidenced by greater seedling density and species richness of forbs and native species in fescue prairie. The higher soil pH value by ash might be the reason for this response, which opens the possibility for future study to explore the mechanisms behind fire in shaping species composition. However, the role of smoke in seedling emergence from the soil seed bank remains unknown, even though seed germination of selected species has been shown to improve by previous studies in fescue prairie. The lack of positive effects of smoke solutions on seedling emergence in the current study may be due to the high concentration.

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